

Practical No: - 2

ON

Cryptography and Network Security - SSCS 4020

TITLE: Playfair Cipher (Using Python)

Implementation of Playfair Cipher for Encryption and Decryption Using Python

**BACHLEOR OF SCIENCE IN INFORMATION TECHNOLOGY
(HONOURS.)**

SUBMITTED BY:

Name: Shah Rishabh Alpeshbhai

Enrolment: 23SS02IT181

Class: BSCIT-H – 7A - Batch 2023-27

Problem Statement – 1: Encryption and Decryption Using Playfair Cipher

Aim: To implement the Playfair Cipher algorithm for encrypting and decrypting a plaintext message using a keyword-based 5×5 matrix.

INPUT:

```
def key_matrix(key):
```

```
    key = "".join(dict.fromkeys(key.upper().replace("J", "I")))
```

```
    chars = key + "".join(c for c in "ABCDEFGHIJKLMNOPQRSTUVWXYZ" if c not in key)
```

```
    return [list(chars[i:i+5]) for i in range(0, 25, 5)]
```

```
def pos(mat, ch):
```

```
    ch = "I" if ch == "J" else ch
```

```
    for i in range(5):
```

```
        for j in range(5):
```

```
            if mat[i][j] == ch:
```

```
                return i, j
```

```
def prepare(text):
```

```
    text = "".join(c for c in text.upper().replace("J", "I") if c.isalpha())
```

```
    res, i = "", 0
```

```
    while i < len(text):
```

```
        a = text[i]
```

```
        b = text[i+1] if i+1 < len(text) else "X"
```

```
        if a == b:
```

```
            res += a + "X"
```

```
            i += 1
```

```
        else:
```

```
            res += a + b
```

```
            i += 2
```

```
    return res + ("X" if len(res) % 2 else "")
```

```
def playfair(text, mat, mode):  
    text = prepare(text) if mode == 1 else text  
    result = ""  
    for i in range(0, len(text), 2):  
        a, b = text[i], text[i+1]  
        r1, c1 = pos(mat, a)  
        r2, c2 = pos(mat, b)  
  
        if r1 == r2:  
            result += mat[r1][(c1+mode)%5] + mat[r2][(c2+mode)%5]  
        elif c1 == c2:  
            result += mat[(r1+mode)%5][c1] + mat[(r2+mode)%5][c2]  
        else:  
            result += mat[r1][c2] + mat[r2][c1]  
    return result  
  
key = input("Enter Keyword: ")  
text = input("Enter Plaintext: ")  
  
mat = key_matrix(key)  
  
print("\nKey Matrix:")  
for row in mat:  
    print(*row)  
  
cipher = playfair(text, mat, 1)  
plain = playfair(cipher, mat, -1)  
  
print("\nEncrypted Text:", cipher)  
print("Decrypted Text:", plain)
```

OUTPUT:

```
Enter Keyword: Rishabh
Enter Plaintext: shah
```

```
Key Matrix:
```

```
R I S H A
B C D E F
G K L M N
O P Q T U
V W X Y Z
```

```
Encrypted Text: HARA
```

```
Decrypted Text: SHAH
```

Problem Statement – 2: Secure Communication Using Playfair Cipher

Aim: To simulate secure message transmission using the Playfair Cipher with a user-defined keyword.

INPUT:

Use the same Playfair Cipher functions from Problem Statement - 1

```
keyword = input("Enter Shared Keyword: ")
```

```
message = input("Enter Secret Message: ")
```

```
matrix = generate_key_matrix(keyword)
```

```
ciphertext = encrypt_playfair(message, matrix)
```

```
print("\nSender Side")
```

```
print("Original Message :", message)
```

```
print("Encrypted Message:", ciphertext)
```

```
received_text = decrypt_playfair(ciphertext, matrix)
```

```
print("\nReceiver Side")
```

```
print("Received Ciphertext:", ciphertext)
```

```
print("Decrypted Message :", received_text)
```

OUTPUT:

```
Enter Shared Keyword: Network Security
Enter Secret Message: Cryptography
```

Sender Side

Original Message : Cryptography

Encrypted Message: UKIQWNFKIVGA

Receiver Side

Received Ciphertext: UKIQWNFKIVGA

Decrypted Message : CRYPTOGRAPHY